Composing layers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

TEACHER GUIDE · 1 OF 9

Prepare the purpose, prerequisites and materials

> Prepare Composing layers: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Trace how one **layer** supplies the next **layer input**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Activation functions

Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

Look for: Trace identity ReLU and sigmoid on a numerical score

Reference example, not the only acceptable wording: Teacher explanation: Identity gives -2, 0, 2; ReLU gives 0, 0, 2. ReLU retains the positive score itself; it is not a step classifier.

Dot products and weighted sums

Without opening the old lesson, explain one key idea from Dot products and weighted sums. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate a dot product by matching feature positions

Reference example, not the only acceptable wording: Teacher explanation: Contributions are 6 and -5, so the score is 1.

Teach **composition** through explicit numerical traces in two 45-minute sessions. Keep the **layer** definition, order and activation visible.

The aim is to understand which values each computation receives, not to memorise a network drawing or claim successful **training**.

Outcomes

Trace how one **layer** supplies the next **layer input**s

Check that connected **layer** dimensions and feature order agree

Explain why intermediate activations are not automatically class **prediction**s

Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Prerequisite answer

ReLU returns 0 and 3. It transforms a numerical score. A class **prediction** requires a separate declared **output** rule.

End goal for the pictorial case

By the end, I can explain composing **layer**s using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain and model the mechanism

> Model Composing layers using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Explain Composing layers · 1

A network connects **layer**s so that the **output** vector of one **layer** becomes the **input** vector of the next. Use a fictional numerical **input** x = [x1, x2].

The first **layer** computes z1 = x1 - x2 and z2 = x1 + x2 - 2, then applies ReLU separately: h1 = max(0, z1), h2 = max(0, z2). Its **output** is the ordered vector h = [h1, h2].

Explain Composing layers · 2

The next **layer** computes y = 2h1 - h2 + 1. It receives h, not the original x. It uses weights [2, -1] and bias 1, with identity as its **output** activation.

This is a two-**layer** computation when counting the **layer**s with weights; an **input** placeholder is not counted here. All parameters are supplied, so this lesson performs no **training**.

The two first-**layer** units both receive the same original x. They can be calculated in either order. Only after both activations are available do we evaluate the **output layer**.

Explain Composing layers · 3

Take x = [3, 1]. First compute z1 = 3 - 1 = 2 and z2 = 3 + 1 - 2 = 2. ReLU gives h = [2, 2]. The **output layer** receives exactly this ordered pair and returns y = 2 times 2 - 2 + 1 = 3.

Record z, h and y separately so that a correct final number cannot hide an incorrect intermediate step.

The code assumes numeric **input**s with the agreed meaning. Its length check does not validate units or field order. Python indices 0 and 1 correspond to the first and second hidden components.

There is no feedback from y into h during this **forward calculation**.

Term	Meaning
Layer	A group of computations that maps an input vector to an output vector
Hidden activation	An intermediate value computed inside a network
Composition	Applying one function to the output of another
Forward calculation	Computing successive values from an input using fixed parameters

Complete each stage before using its output

AVAILABLE INPUT	COMPUTATION	VALUE PASSED ON
x is [3, 1]	Both first layer scores	z is [2, 2]
z is [2, 2]	ReLU on each component	h is [2, 2]
h is [2, 2]	2h1 - h2 + 1	y is 3

Read the rows from top to bottom. The first two rows together form the hidden layer; the final row is the output layer. Each row shows the values actually passed to the next stage.

Explain Composing layers · 4

Guided: x = [1, 1] gives z = [0, 0], h = [0, 0], y = 1. Using x directly would give 2, skipping the hidden transformation.

Exit: compatible shapes and execution check whether the specified computation can run, not whether **prediction**s match real labels. Accuracy requires appropriate labelled examples, a metric and independent evaluation.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Complete each stage before using its output

AVAILABLE INPUT	COMPUTATION	VALUE PASSED ON
x is [3, 1]	Both first layer scores	z is [2, 2]
z is [2, 2]	ReLU on each component	h is [2, 2]
h is [2, 2]	2h1 - h2 + 1	y is 3

Read the rows from top to bottom. The first two rows together form the hidden layer; the final row is the output layer. Each row shows the values actually passed to the next stage.

Trace x = [1, 1]. Record z1, z2, h1, h2 and y. Explain why using x directly in the output formula would be a different model.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Composing layers

Compose two hidden detectors before one **output** instead of skipping a **layer**.

The facts we will use

> **Supplied facts**
>
> For **input** [u, v], h1=ReLU(u-2), h2=ReLU(v-1), y=h1-h2. **Input** [4, 3].

> **Supplied facts**
>
> All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/5

Composing layers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. First hidden unit

u-2=2, so h1=2.



2. Second hidden unit

v-1=2, so h2=2.



3. Compose output

y=2-2=0; the output uses activated hidden values, not raw inputs.

CHANGE ONE THING

Change only v from 3 to1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. First hidden unit	u-2=2, so h1=2.
2. Second hidden unit	v-1=2, so h2=2.
3. Compose output	y=2-2=0; the output uses activated hidden values, not raw inputs.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Composition means one **layer**'s result becomes the next **layer**'s **input**. Compute both hidden units before combining them. At [4,3], their equal **output**s cancel.

Lowering only the second **input** removes its positive hidden contribution, so subtracting it no longer reduces the **output**. The final subtraction must use activated hidden values rather than the raw measurements.

What this example does not establish

Keep the stated limit

Hand-composed detectors illustrate **layer** order; they are not **evidence** of a learned or biologically realistic neural **mechanism**.

Change one thing in the pictured case

Change one thing

Change: Change only v from 3 to1.

Predicted result

Prediction: h1 stays 2, h2 becomes 0, y becomes 2.

Why the result follows

Reason: The second hidden unit’s preactivation reaches 0 while the first unit is unchanged.

Ask while showing the picture

Ask while showing the picture · 5

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Composing layers, reduce support deliberately and check individual reasoning before discussing answers.

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve ReLU at negative, zero and positive scores. Ask students to distinguish an activation from a class.
7 to 18 minutes	Write both first-layer formulas side by side. Draw arrows from the same x to both; label the ordered hidden output.
18 to 32 minutes	Trace [3, 1] together, then use W1. Require scores before ReLU and output contributions after ReLU.
32 to 45 minutes	Run or trace the optional Python and complete the guided [1, 1] case. Ask why directly substituting x into the output formula is incorrect.
Session 2 start to 8 minutes	Retrieve the two layer input/output sizes. Let students describe composition in one sentence.
8 to 20 minutes	Complete W2 and W3. Compare a swap-insensitive case with a revealing case before discussing bug coverage.
20 to 35 minutes	Collect W4 independently before hints. Check both pre-activation scores, both activations, q and class under the declared threshold.
35 to 45 minutes	Complete W5 and the exit response. Give targeted feedback and the fresh retry without replacing the original assessment record.

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide a five-column trace table labelled x, z, h, **output** contributions and y. Allow arithmetic support while requiring the student to name each connection.

Extension: replace hidden ReLU with identity and simplify the lesson network to y = x1 - 3x2 + 3. Test [1, 3] to show that the simplified affine model differs from the ReLU model.

Reduce help deliberately

Completion practice

Completion task: A revised hidden **layer** returns [h1, h2, h3]. The **output layer** still has weights [2, -1]. Explain the mismatch and what must be specified before calculation.

Reasoning to listen for

Teacher check: Teacher explanation: Three hidden values need three matching weights under this dense **output** definition. Specify all three coefficients and the bias or explicitly design a documented selection operation.

Do not silently drop the third value.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Composing **layer**s. Proposed explanation: “The second hidden unit receives the first hidden unit **output**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Both units in this hidden **layer** receive x. The next **layer** receives the pair of activations after both are computed.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

What intermediate value explains the changed **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

h2 changes from 2 to 0 because ReLU(1-1)=0; subtracting 0 rather than 2 raises y from 0 to2. Changed-case check: h1 stays 2, h2 becomes 0, y becomes 2.

The second hidden unit's preactivation reaches 0 while the first unit is unchanged.

Changed-case reasoning: h1 stays 2, h2 becomes 0, y becomes 2. The second hidden unit's preactivation reaches 0 while the first unit is unchanged.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the investigation and diagnose misconceptions

> Run the Composing layers investigation and use the first unsupported step to choose a teaching response.

The second hidden unit receives the first hidden unit **output**: Both units in this hidden **layer** receive x. The next **layer** receives the pair of activations after both are computed.

Two numbers mean the connection must be correct: Matching lengths do not establish matching order or meaning. Trace a case with unequal **hidden activation**s.

A hidden unit is already a class **prediction**: Its activation is an intermediate numerical feature. A declared **output** rule assigns the final class.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Composing layers questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

Guided: x = [1, 1] gives z = [0, 0], h = [0, 0], y = 1. Using x directly would give 2, skipping the hidden transformation.

Exit: compatible shapes and execution check whether the specified computation can run, not whether **prediction**s match real labels. Accuracy requires appropriate labelled examples, a metric and independent evaluation.

W1 Trace the second worked case

Workbook question · W1

For x = [1, 3], write both scores, both ReLU activations and y using the lesson model.

Teacher reasoning · W1

Teacher explanation: Scores are -2 and 2; activations are 0 and 2; y = 0 - 2 + 1 = -1.

W2 Repair a dimension mismatch

Workbook question · W2

A revised hidden **layer** returns [h1, h2, h3]. The **output layer** still has weights [2, -1]. Explain the mismatch and what must be specified before calculation.

Teacher reasoning · W2

Teacher explanation: Three hidden values need three matching weights under this dense **output** definition. Specify all three coefficients and the bias or explicitly design a documented selection operation.

Do not silently drop the third value.

W3 Use a revealing order check

Workbook question · W3

Compare hidden vectors for x = [4, 1] and x = [3, 0]. Which **input** reveals a bug that swaps h1 and h2 but keeps **output** weights unchanged? Give correct and swapped **output**s for that **input**.

Teacher reasoning · W3

Teacher explanation: The first gives [3, 3] and cannot expose a swap. The second gives [3, 1]; correct **output** 6 becomes 0 when values alone are swapped.

W4 Compose a new network independently

Workbook question · W4

A new model takes [u, v]. Compute a = ReLU(u + v - 3), b = ReLU(u - v), then q = a + 2b - 1. Trace both **input**s [2, 1] and [1, 2], including scores before ReLU. Apply class 1 when q >= 0.

State the hidden and **output** vector sizes.

Teacher reasoning · W4

Teacher explanation: For [2, 1], scores 0 and 1 give [a, b] = [0, 1], q = 1 and class 1. For [1, 2], scores 0 and -1 give [0, 0], q = -1 and class 0. Hidden dimension is 2; **output** dimension is 1.

The equality at hidden score zero produces activation zero, not an automatic final class.

W5 Explain information loss

Workbook question · W5

Use the lesson model to verify h and y for [0, 1] and [0, 2]. Could a fixed later **layer** receiving only h distinguish the two original **input**s?

Teacher reasoning · W5

Teacher explanation: Scores are [-1, -1] and [-2, 0]. Both produce h = [0, 0] and y = 1.

A deterministic later computation given the same h produces the same result; additional original-**input** information would require a changed architecture.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

W4 requires both complete traces, correct classes and sizes. Fresh retry: a = ReLU(u - 1), b = ReLU(v - 2), q = 2a - b.

Inputs [0, 3] and [2, 2] give pre-activation scores [-1, 1] and [1, 0], hidden vectors [0, 1] and [1, 0], and q values -1 and 2. Under q >= 0 their classes are 0 and 1. Ask which vector the **output layer** actually receives.

Assess independent transfer

Independent question · withhold the key

Selected case: A new model takes [u, v]. Compute a = ReLU(u + v - 3), b = ReLU(u - v), then q = a + 2b - 1. Trace both **input**s [2, 1] and [1, 2], including scores before ReLU. Apply class 1 when q >= 0.

State the hidden and **output** vector sizes.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: For [2, 1], scores 0 and 1 give [a, b] = [0, 1], q = 1 and class 1. For [1, 2], scores 0 and -1 give [0, 0], q = -1 and class 0. Hidden dimension is 2; **output** dimension is 1.

The equality at hidden score zero produces activation zero, not an automatic final class.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect the project, support and next lesson

> Connect Composing layers to Image-pattern Network Investigation, adapt support and plan a later changed-case check.

Add a labelled composing **layer**s **evidence** sheet to Image-pattern Network Investigation. Show how you can trace how one **layer** supplies the next **layer input**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide a five-column trace table labelled x, z, h, **output** contributions and y. Allow arithmetic support while requiring the student to name each connection.

Extension: replace hidden ReLU with identity and simplify the lesson network to y = x1 - 3x2 + 3. Test [1, 3] to show that the simplified affine model differs from the ReLU model.

Week 6 develops forward passes by hand and systematic checking of intermediate values.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/6, ai-5/8, ai-5/11. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Composing layers lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Trace how one **layer** supplies the next **layer input**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: A revised hidden **layer** returns [h1, h2, h3]. The **output layer** still has weights [2, -1]. Explain the mismatch and what must be specified before calculation.

Supported practice answer

Reasoned practice answer: Teacher explanation: Three hidden values need three matching weights under this dense **output** definition.

Specify all three coefficients and the bias or explicitly design a documented selection operation. Do not silently drop the third value.

Check every learner before discussion: Use the worked case for Composing **layer**s. Proposed explanation: “The second hidden unit receives the first hidden unit **output**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Both units in this hidden **layer** receive x. The next **layer** receives the pair of activations after both are computed.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Change only v from 3 to1. Require a **prediction** and a reason before comparing. Expected result: h1 stays 2, h2 becomes 0, y becomes 2.

Reason: The second hidden unit’s preactivation reaches 0 while the first unit is unchanged.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

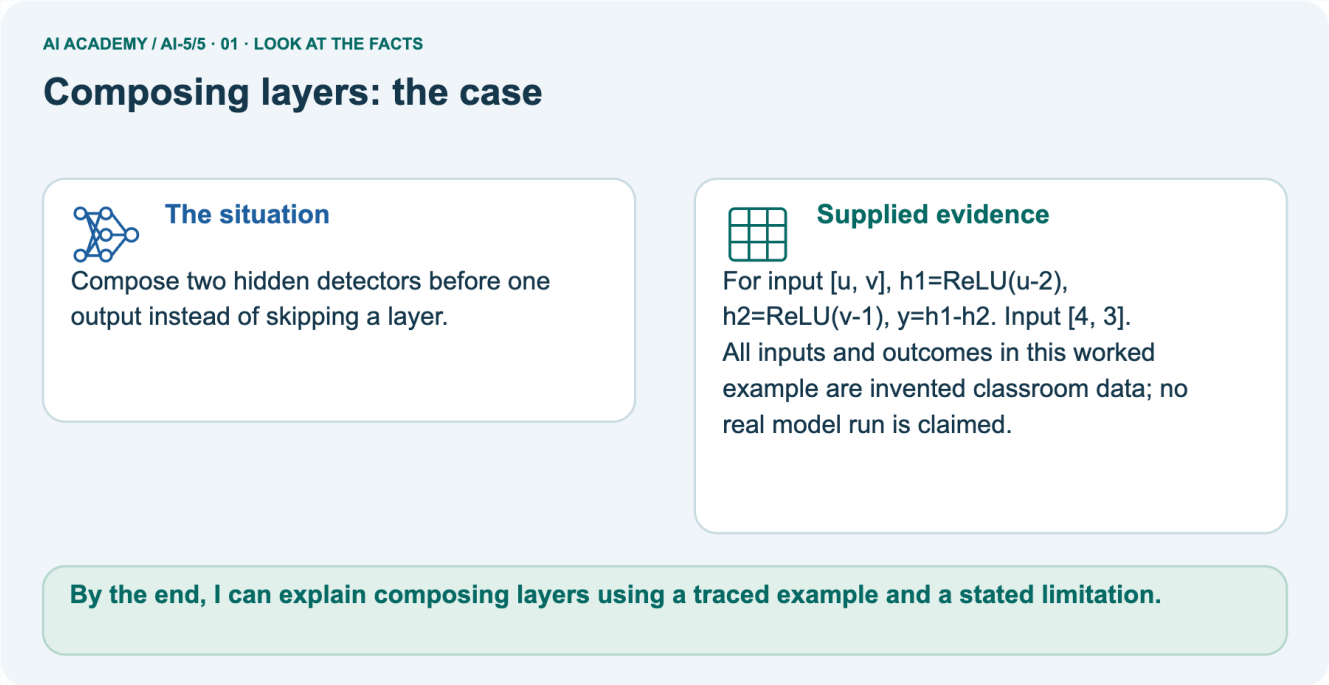
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Composing layers graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Composing layers: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Compose two hidden detectors before one **output** instead of skipping a **layer**. Supplied facts: For **input** [u, v], h1=ReLU(u-2), h2=ReLU(v-1), y=h1-h2. **Input** [4, 3].

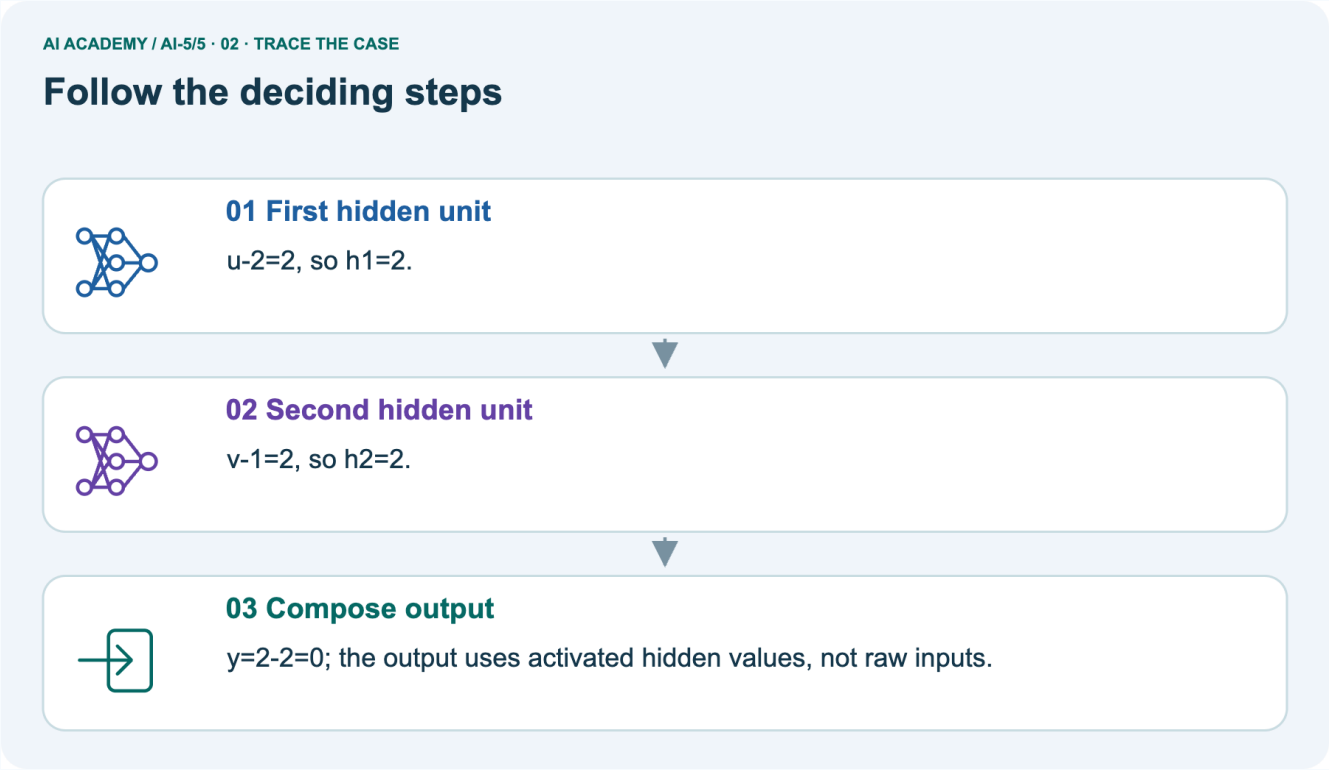
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

First hidden unit: u-2=2, so h1=2. → Second hidden unit: v-1=2, so h2=2. → Compose **output**: y=2-2=0; the **output** uses activated hidden values, not raw **input**s.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Complete each stage before using its output

AI ACADEMY / AI-5/5 · 03 · INSPECT THE LESSON MODEL

Complete each stage before using its output

Available input	Computation	Value passed on
x is [3, 1]	Both first layer scores	z is [2, 2]
z is [2, 2]	ReLU on each component	h is [2, 2]
h is [2, 2]	2h1 - h2 + 1	y is 3

Read the rows from top to bottom. The first two rows together form the hidden layer; the final row is the output layer. Each row shows the values actually passed to the next stage.

Follow the labelled picture with the matching explanation. Open the vector diagram

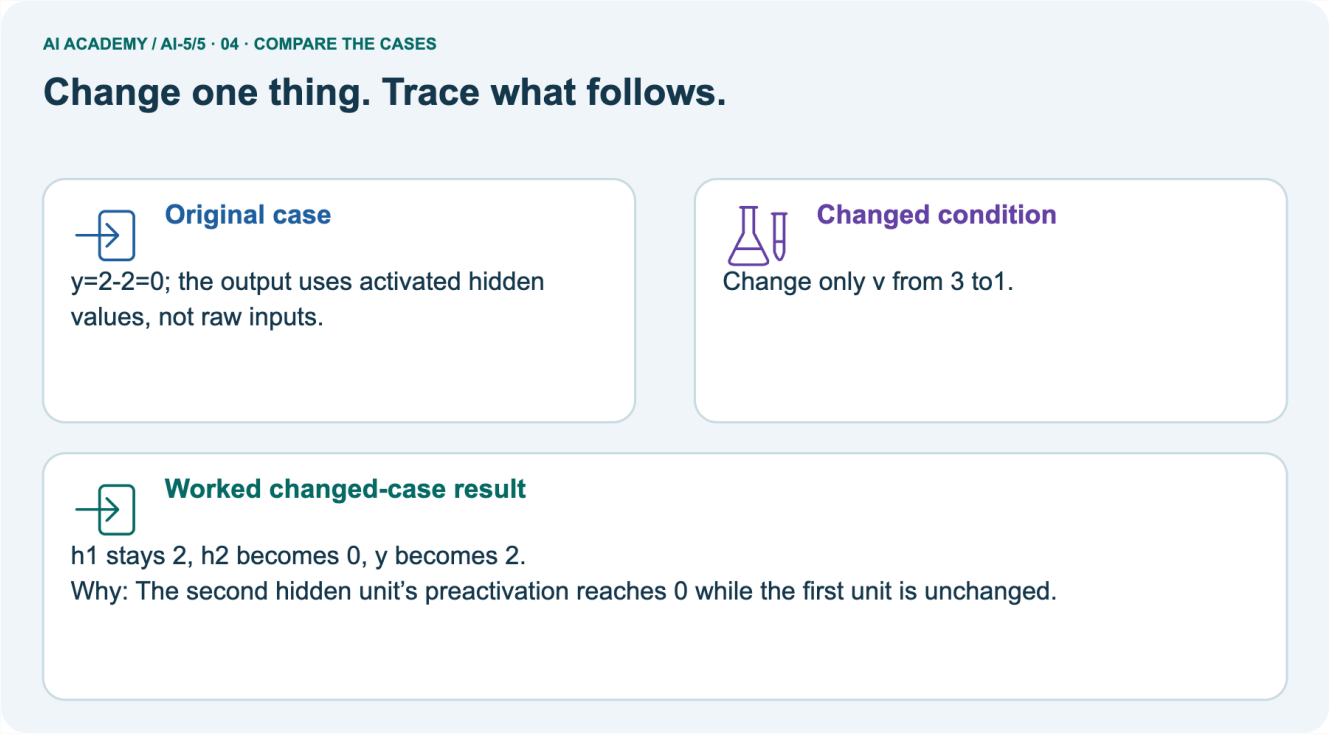
Complete each stage before using its **output**. x is [3, 1] | Both first **layer** scores | z is [2, 2]. z is [2, 2] | ReLU on each component | h is [2, 2]. h is [2, 2] | 2h1 - h2 + 1 | y is 3

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: y=2-2=0; the **output** uses activated hidden values, not raw **input**s. Changed condition: Change only v from 3 to1. Expected result: h1 stays 2, h2 becomes 0, y becomes 2.

Reason: The second hidden unit's preactivation reaches 0 while the first unit is unchanged.

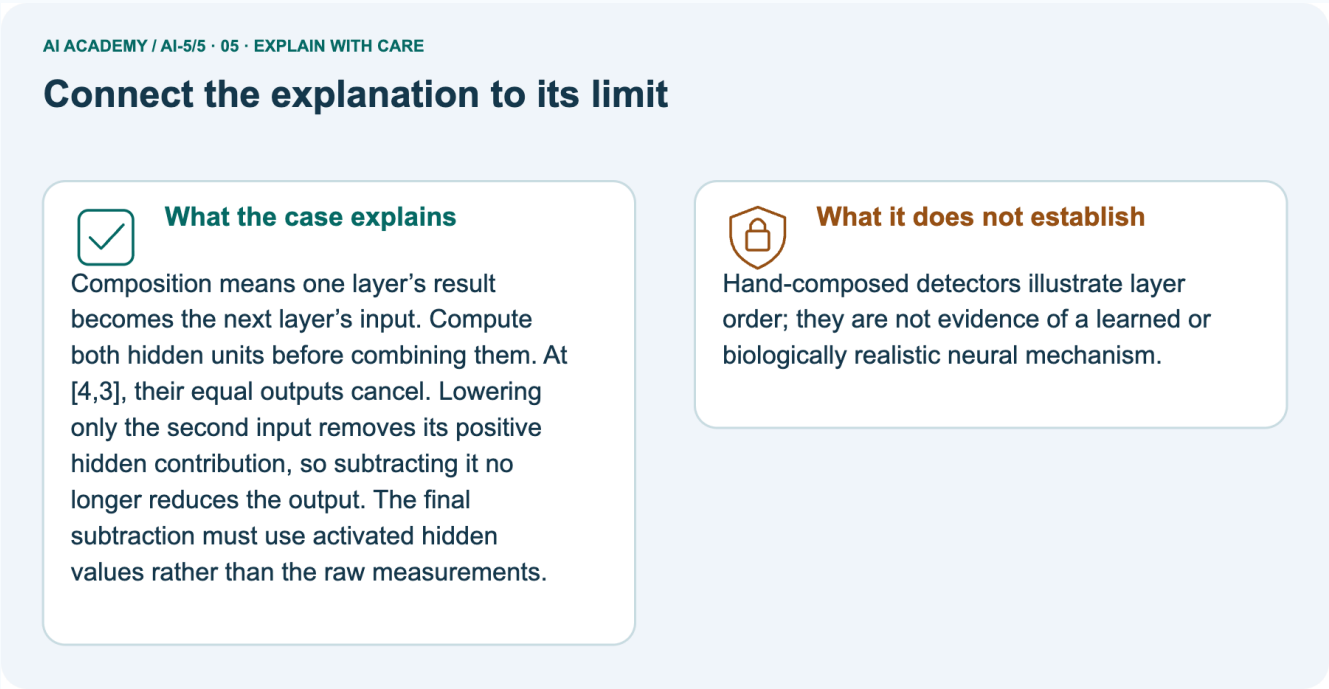
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The second hidden unit's preactivation reaches 0 while the first unit is unchanged.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Composition means one **layer**'s result becomes the next **layer**'s **input**. Compute both hidden units before combining them. At [4,3], their equal **output**s cancel.

Lowering only the second **input** removes its positive hidden contribution, so subtracting it no longer reduces the **output**. The final subtraction must use activated hidden values rather than the raw measurements.

Limit: Hand-composed detectors illustrate **layer** order; they are not **evidence** of a learned or biologically realistic neural **mechanism**.

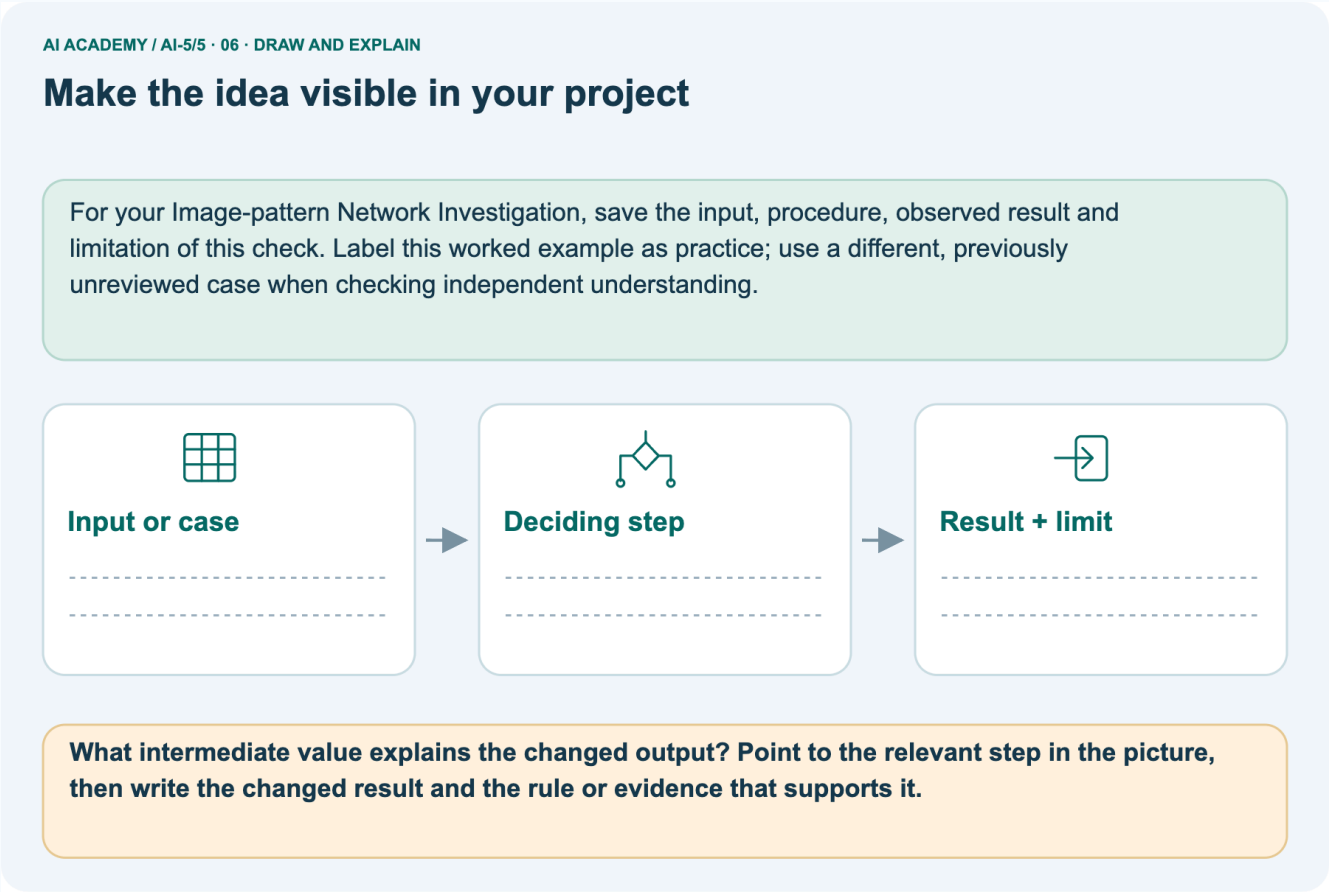
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Hand-composed detectors illustrate **layer** order; they are not **evidence** of a learned or biologically realistic neural **mechanism**.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What intermediate value explains the changed **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: h2 changes from 2 to 0 because ReLU(1-1)=0; subtracting 0 rather than 2 raises y from 0 to2. Changed-case check: h1 stays 2, h2 becomes 0, y becomes 2.

The second hidden unit’s preactivation reaches 0 while the first unit is unchanged.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.